

Yanjin Li

Self-taught AI Engineer & Researcher — research · systems · applied ML · product

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Self-taught AI builder — learning from papers and courses since early 2026, I designed, built and evaluated **11 projects in three months**, spanning training-method research, multi-device AI systems, applied decision systems and shipped products.

SELECTED PROJECTS

Zero Gradient — 4.16B-parameter LM trained with no backpropagation RESEARCH

Trained a content-routed MoE language model with autograd globally disabled on a single T4; ran a 5-phase capability-boundary study and a controlled experiment proving the ceiling is architectural, not algorithmic. Paper draft + arXiv package.

[PyTorch](#) · [MoE](#) · [layer-local learning](#) · [WikiText-103](#)

Gemma4all — cross-device AI task system SYSTEMS

Schema-first (42 JSON schemas), event-sourced, local-first system with fail-closed human-in-the-loop approvals and a privacy-based routing engine across iOS, Android and macOS; on-device inference via LiteRT-LM on Metal GPU.

[TypeScript](#) · [Python](#) · [Swift](#) · [Kotlin](#) · [JSON Schema](#) · [event sourcing](#)

TriageGeist — trustworthy emergency-triage decision stack APPLIED ML

Found 99.7% label leakage and refused to exploit it; built a text-blind GBDT (QWK 0.93) wrapped in isotonic calibration, conformal uncertainty, a bootstrap fairness audit and an operations console. 12/12 tests.

[LightGBM](#) · [conformal prediction](#) · [calibration](#) · [fairness audit](#)

NeuroGolf (ARC-AGI) — minimal ONNX circuit compiler RESEARCH

Reframed an ARC-AGI "neural golf" contest as program synthesis; reverse-engineered the scorer into a cost model and compiled true task rules into overfitting-proof integer-weight circuits validated on 3000+ generated samples.

[ONNX](#) · [program synthesis](#) · [scorer reverse-engineering](#)

AI Showrunner — one sentence → subtitled vertical short film PRODUCT

Multimodal agent pipeline (script → storyboard → video generation) with a vision-model critic loop driving targeted retries and cross-shot character consistency; six end-to-end films produced.

[FastAPI](#) · [Qwen-VL critic](#) · [I2V/T2V](#) · [ffmpeg](#)

Nemotron Post-training — post-training pipeline for a 30B MoE reasoning model RESEARCH

Built a reproducible rule-teacher → CoT-SFT pipeline (baseline EM 0.75; 30B MoE loaded in 4-bit on one GPU, upstream cache bug found and patched, multi-dimensional reward shaping), then root-caused the training failure across six postmortems and designed the recovery route.

[Transformers](#) · [PEFT/LoRA](#) · [4-bit quantization](#) · [GRPO reward shaping](#)

Also built: Offline French Companion (in-browser LLM tutor — WebLLM + Whisper + OCR, fully offline) · Orbit Wars (multi-agent evaluation methodology) · Maze Crawler (strategy agent with CI) · VoiceInput (pure-Swift macOS speech input) · Atelier Post (shipped art-to-Instagram tool, web/PWA/iOS).

PROFESSIONAL EXPERIENCE

BNP Paribas — Global Corporate Banking Intern

Paris · Dec 2025 — Present

- Structured tailored cross-border transactional and financing solutions for multinational clients; built and presented pitchbooks combining market analysis with pricing and cross-border liquidity.
- Helped land internal AI tooling — built RAG knowledge bases on Open WebUI and Dify to improve client-response efficiency (book incl. CATL, BYD, Alipay, Midea, TCL, JD.com).

HSBC — Global Markets, Liquidity & Investment Intern

Paris · Jun — Nov 2024

- Monitored Fed/ECB/BoE rates and money markets to optimize short-term EMEA portfolios under LCR/NSFR; built automated VBA/Excel tools tracking rate moves and pricing gaps in real time.
- Developed dynamic pricing models for wholesale deposits; contributed to the ALCM committee (gap-risk analysis, funding-cost optimization).

TikTok — Growth Strategy Intern

Beijing · Feb — May 2023

- Contributed to 10+ strategic projects, growing SaaS/AI market share by 23%; ran SQL/Excel analytics for a 20+ member team and lifted user conversion by 13% in one month.

Uber China · DiDi — Product Management & User Operations Intern

Beijing · Aug 2022 — Jan 2023

- Managed 3 business-operations projects; analyzed large datasets in Python, SQL and VBA to build predictive models and data-driven user-interaction strategies.

EDUCATION

ESCP Business School (Paris & London) — Master in Management, Finance2023 — 2026

Beijing International Studies University — BA Language & Culture (French, Japanese); minor New Media (CUC); self-taught Psychology (Yale/Coursera)

2019 — 2023

SKILLS

Programming: Python · TypeScript/Node · Swift · Kotlin · JavaScript · SQL · VBA

AI / ML: PyTorch · LoRA/PEFT · MoE · calibration · conformal prediction · LiteRT-LM · ONNX · WebLLM · Whisper · LLM/agent orchestration · schema-first & event-sourced systems

Spoken languages: Chinese (native) · English (fluent) · French (fluent) · Japanese (working)

Strengths: High agency and a self-taught, zero-to-one builder — curious, reflective and resilient, with strong insight and a painter's eye for design, at home across cultures and languages.